

Switching the loop on at inference

cuts E rate $\approx 15\times$

exp038 · 30 May 2026

The trained networks this entry uses are produced once in the shared training hub, exp022 (Training), and reused here rather than retrained. Its complete TR-02 activity frontier contains seeds 42–44 at every COBA and PING rate target. The machine-readable results retain each checkpoint row and report the across-seed mean and SEM for every frontier point.

Abstract

Loads a trained COBA network and switches the I-loop on at inference by sweeping $ei_strength$ from 0 to 1. Across independently trained seeds 42–44, the same feedforward weights that fire at ≈ 133 Hz without the loop fire at ≈ 9 Hz with it engaged, a $\approx 15\times$ drop with no weight update. Accuracy, though, falls ≈ 36 pp (from $\approx 90\%$ to $\approx 55\%$) as the loop engages, because the readout was never trained with it. PING gating is a post-hoc sparsity knob: the architecture, not the training, supplies the gamma dynamics, but using it well still needs training *with* the loop.

Method

Each inference intervention starts from the checkpoint selected by validation accuracy. The recurrent-loop replacements and input-rate sweeps therefore probe the selected classifier, not the final-epoch training state.

Parameter	Value
Integration timestep Δt	0.1 ms
Trial duration T	200 ms
Evaluation corpus	Official MNIST test partition (10000 samples)
Evaluated held-out samples per seed	1000
Epochs	50

The PING and COBA baseline definitions and the training recipe are in exp025 this entry runs the COBA $\rightarrow$ PING I-loop transfer probe at eval time on the trained baselines. (The input-rate sweep / f-I curve material that previously lived here has moved to exp023, the natural home for “architectural response to drive”.)

Inference-time probe. The trained COBA baselines (seeds 42–44, activity regulariser off) are loaded and $ei_strength$ (the I-loop gain) is overridden at eval time across 11 values from 0 to 1. Each seed uses the same grid and held-out sample count. W_{in} and W_{out} load from its COBA checkpoint; W^{EI} and W^{IE} are freshly initialised (the COBA checkpoint stores these

at zero, so skipping the load leaves a functional I-loop). No retraining. The quantitative curves report the across-seed mean and shaded sample SD. The raster panels show seed 42 as an explicitly illustrative example.

Results

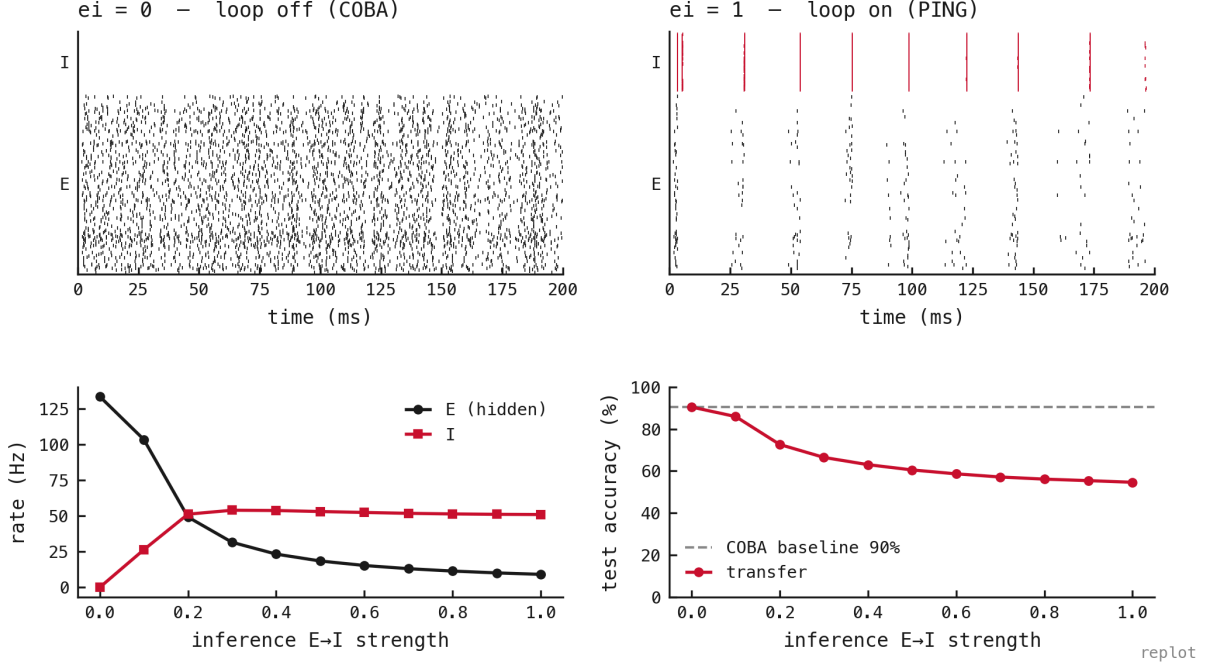


Figure 1: Switching the recurrent I-loop on *at inference* on a trained COBA network (no retraining; W^{EI}/W^{IE} freshly wired, since the COBA checkpoint stores them at zero). **Top:** the same feedforward weights fire densely and asynchronously at $ei = 0$ (COBA) and in gamma bands at $ei = 1$ (PING); the gamma dynamics come from the inhibitory architecture, not from training. **Bottom left:** across-seed mean E rate falls $\approx 15\times$ ($\approx 133 \rightarrow 9$ Hz) as the loop engages while I rises to ≈ 51 Hz; the suppression is continuous in loop strength. **Bottom right:** mean accuracy *degrades* without retraining, from the $\approx 90\%$ COBA baseline to $\approx 55\%$ at full strength (a ≈ 36 pp cost). Shaded bands show sample SD across seeds 42–44. So the architecture supplies the rate-gating for free, but using it well needs training *with* the loop (exp025): the sparsity is architectural, the accuracy is learned.

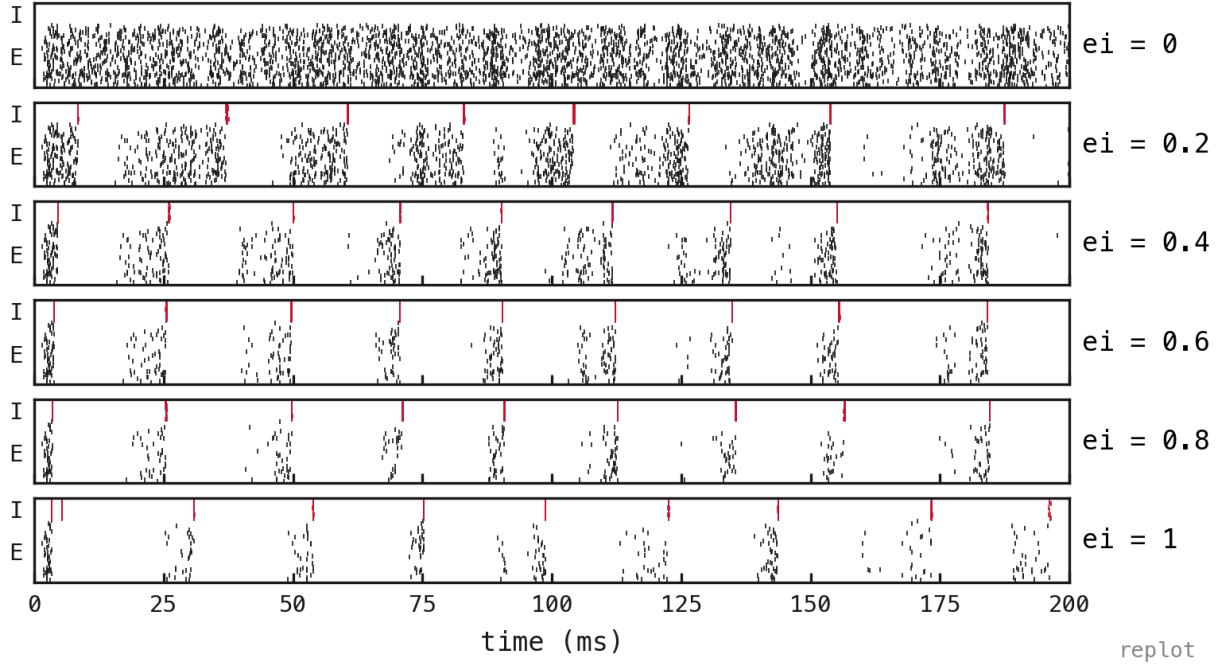


Figure 2: The illustrative seed-42 transition: trained COBA replayed at six inference-time $ei_strength$ values (same trial, same feedforward weights, a fresh I-loop each row). At $ei = 0$ the asynchronous-dense COBA pattern persists; by $ei \approx 0.4$ the same weights produce gamma cycles, sharpening toward $ei = 1$. The rhythm appears continuously as the loop is wired in, with no retraining at any point.